

Lab 1: Introduction to Python & Backpropagation using Python

Question: You have recently joined an AI startup that is developing an intelligent quality inspection system for a manufacturing plant. The system must classify products as **Acceptable (1)** or **Defective (0)** based on sensor measurements.

Your team has decided to implement a simple **Artificial Neural Network (ANN)** with one hidden layer and train it using the **Backpropagation algorithm** in Python.

As a Machine Learning Engineer, your task is to implement the neural network from scratch (without using machine learning libraries such as TensorFlow, PyTorch, or Scikit-learn) and analyze its learning behaviour.

Tasks

Part A – Implementation (60 Marks)

Develop a Python program that performs the following:

1. Import the required Python libraries.
2. Define the input dataset and target outputs.
3. Initialize weights and biases randomly.
4. Implement:
 - Forward propagation
 - Sigmoid activation function
 - Error calculation
 - Backpropagation
 - Weight updates using Gradient Descent
5. Train the network for **5000 epochs**.
6. Display:
 - Initial weights
 - Final weights
 - Training error
 - Predicted outputs

Part B – Observation & Analysis (40 Marks)

After successfully executing the program, answer the following questions.

Q1. What is the purpose of the **forward propagation** step?

Q2. Why is the **Sigmoid activation function** used in this experiment?

Q3. Record the **initial** and **final** loss values. What do these values indicate about the learning process?

Q4. Why is **Backpropagation** considered the learning mechanism of an Artificial Neural Network?

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Q5. What would happen if the **learning rate** is increased from **0.1** to **1.0**? Explain your observation.

Q6. Train the model using **100**, **1000**, and **5000** epochs. Complete the following table.

Epochs	Final Error	Prediction Accuracy
100		
1000		
5000		

Comment on the relationship between the number of epochs and the training performance.

Q7. Suppose the hidden layer contains **10 neurons instead of 4**. Will the network always perform better? Justify your answer.

Q8. Why are the weights initialized randomly instead of assigning all weights as zero?

Q9. Explain the role of the learning rate in Backpropagation. What problems arise if the learning rate is:

- Too small?
- Too large?

Q10. List any **three real-world applications** where Backpropagation-based neural networks are used.

Critical Thinking Question

Modify the program to replace the **Sigmoid activation function** with **ReLU** in the hidden layer. Compare:

- Number of epochs required for convergence
- Final loss
- Predicted outputs

Discuss whether ReLU improves the performance for this problem.

Expected Learning Outcomes

After completing this experiment, students should be able to:

- Develop Python programs for Artificial Neural Networks.
- Implement the Backpropagation algorithm from scratch.
- Understand forward and backward propagation.
- Analyze the effect of learning rate and epochs.
- Interpret neural network training results.
- Relate ANN concepts to real-world AI applications.